Design and Fabrication of Microfluidics Devices using Soft Lithography

By

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Allah is the guardian of those who believe. He brings them out of the darkness into the light; and (as to) those who disbelieve, their guardians are Shaitans who take them out of the light into the darkness; they are the inmates of the fire, in it they shall abide.

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Dedication

To the unwavering pillars of my academic odyssey, my parents, teachers, and mates—your collective influence has been the driving force behind this thesis. My parents, your boundless love and unwavering support have shaped the bedrock of my aspirations; to my teachers, your impassioned guidance has transcended textbooks, leaving an indelible mark on both my academic pursuits and character; and to my mates, in the collaborative tapestry of shared challenges and triumphs, your camaraderie has been an invaluable source of strength. This thesis stands as a testament to the sacrifices, wisdom, and collective spirit that have propelled me through the intricate realms of knowledge and personal growth. With profound gratitude, I dedicate this work to you all, recognizing the profound impact each of you has had on my academic journey.

Table of Contents

Chapter 1 Introduction 9
1.1 Microfluidics 9
1.1.1 Microfluidics Applications 9
1.1.2 PDMS Microfluidics 9
1.2 Problem Statement 12
CHAPTER 2 Literature Review 13
CHAPTER 3 Design Methodology 15
3.1 Problem Statement 15
3.2 CAD MODEL 16
3.3 Microfabrication Technique 17
Chapter 4: Fabrication Methodology 20
4.1 Preparation of Mold 20
4.1.1 Argon Plasma Treatment 20
4.1.2 Spin Coating with Photoresist 21
4.1.3 Soft Baking 23
4.1.4 UV Exposure 24
4.1.5 Post-Exposure Baking 25
4.1.6 Pattern Visibility Check 26
4.1.7 Developing the Channel 26
4.2 PDMS Casting 27
4.2.1 Mixing PDMS Elastomer and Curing Agent 27
4.2.2 Degassing the Mixture 29
4.2.3 PDMS Casting and Curing 30
4.2.4 Punching Holes 31
4.2.5 Plasma Bonding 31
4.3 Validation and Testing 32
4.4 Conclusion 32

List of Figures

Figure 1:Scheme describing rapid prototyping of microfluidic systems 11
Figure 2: CAD Model 16
Figure 3: Engineering Drawing of Previous Model 16
Figure 4: Overview of preparation of mold using Photolithography 17
Figure 5: PDMS Casting Process 18
Figure 6: Surface Wetting (PDMS) 19
Figure 7: Plasma Treatment 20
Figure 8: Water droplet on a plasma treated surface 21
Figure 9: Water droplet on an untreated surface 21
Figure 10: Extracting value of speed against desired thickness 22
Figure 11 : Coating substrate with SU-8 photoresist 22
Figure 12 : Soft Baking 23
Figure 13 : UV Exposure 24
Figure 14 : Photomask 24
Figure 15 : Placing substrate in oven for PEB 26
Figure 16: Pattern Visibility Check 26
Figure 17 : Developing the Channel 27
Figure 18 : Weighing PDMS Elastomer 28
Figure 19 : PDMS+Curing Agent weight 28
Figure 20: Mixture Containing Bubbles 29
Figure 21 : Degassing through Vacuum Desiccator 29
Figure 22 : Vacuum Chamber 29
Figure 23 : Pouring PDMS 30
Figure 24 : Curing PDMS 30
Figure 25: Peeling of f the PDMS 30
Figure 26: Punching Holes 31
Figure 27 : Plasma Bonding 31

List of Tables

Table 1: Properties of PDMS [5] 10
Table 2: Table of Diffusion Coefficients in Water 15
Table 3 : Soft Baking Time Duration for Required Thickness 23
Table 4 : UV Exposure Energy against required thickness 25
Table 5: Post Exposure Baking Time 25
Table 6 : Development time against required thickness 27

Chapter 1 Introduction

There has been a significant increase in the attention of microfluidic devices since their introduction in 1990s. Miniaturization, highly throughout precise and control, reduced reagent consumption and improved sensitivity have made these devices very appealing for numerous biological and chemical applications. Microfluidic devices exhibit significant potential for decreasing manufacturing costs, reagent consumption, and analysis time, while concurrently enhancing device efficiency and portability. [1]

1.1 Microfluidics

Microfluidic systems typically come with channels that are usually tens to hundreds of microns in size. These systems manage small amounts of fluids, ranging from 1 atto-liter to 1 nano-liter. [2,3]In the beginning, silicon, quartz or glass materials were used for fabrication of microfluidic devices. Most research in microfluidic system is now carried out in polymers. Fabrication in polymers is easier, more flexible, and much less expensive than in silicon or glass.

1.1.1 Microfluidics Applications

The micro-nanometer sized droplets have enormous applications in cosmetics, food, printing, nutrition and pharmaceutical industry, as a micro-reactors for chemical reactions, as drug carriers, in bio-analyses and material synthesis, catalysis, suspensions and for manipulation of samples of a single cell or a single molecule. [4]

1.1.2 PDMS Microfluidics

A famous name in this context is of PDMS polymer. PDMS, or polydimethylsiloxane, has been extensively employed in the research and advancement of microfluidics. PDMS is an elastomer that is optically transparent, and its flexibility can be adjusted, ranging from very soft (easily deformable by finger pressure) to much stiffer. Creating microchannel systems in PDMS is notably uncomplicated. Using PDMS allows for the quick prototyping of devices and simplifies the demonstration and testing of new ideas. The physical and chemical properties of PDMS also enable the construction of devices with a diverse range of functions, extending from molecular analysis to the creation of frequency-tunable lasers.

PDMS, being elastomeric with a tunable Young's modulus of approximately 750 kPa, displays easy deformation, surface conformity, and non-damaging release from molds, even with undercut features. This unique property makes PDMS suitable for high-fidelity replication of microfluidic channels and micron-scale features through replica molding. The use of composite stamps, combining a stiff layer with a flexible PDMS layer, allows replication of features below 1 nm. Additionally, PDMS's elastomeric nature enables the mechanical tuning of optical components. For instance, stretching or compressing PDMS-based surface-relief gratings or Fresnel lenses alters their periodicity, affecting the diffraction pattern or focal properties, showcasing the versatility of PDMS in creating tunable optical devices.

Property	Characteristics	Consequence
Optical	Transparent; UV cutoff, 240 nm	Optical detection from 240 to 1100 nm
Electrical	Insulating; breakdown voltage, 2×10^7 V/m	Allows embedded circuits; intentional breakdown to open connections
Mechanical	Elastomeric; tunable Young's modulus, typical value of ~750 kPa	Conforms to surfaces; allows actuation by reversible deformation; facilitates release from molds
Thermal	Insulating; thermal conductivity, 0.2 W/(m·K); coefficient of thermal expansion, 310 µm/(m·°C); stable up to ~300°C	Can be used to insulate heated solutions; does not allow dissipation of resistive heating from electrophoretic separation
Interfacial	Low surface free energy ~ 20 erg/cm^2	Replicas release easily from molds; can be reversibly sealed to materials; not wetted by water unless oxidized to SiOH presenting surface
Permeability	Low permeability to liquid water; permeable to gases and nonpolar organic solvents	Contains aqueous solutions in channels; allows gas transport through the bulk material; incompatible with many organic solvents
Reactivity	Inert; can be oxidized by exposure to a plasma	Unreactive toward most reagents; surface can be etched; can be modified to be hydrophilic and also reactive toward silanes
Toxicity	Nontoxic	Can be implanted in vivo; supports mammalian cell growth

Table 1: Properties of PDMS [5]

PDMS systems are commonly manufactured using soft lithography techniques. In soft lithography, a topographically defined structure on a master (typically in photoresist) is replicated onto a soft elastomer. This process can be conducted under regular laboratory conditions and allows for multiple replications. Soft lithography offers a quick, uncomplicated and cost-effective fabrication process. The microfluidic channels are initially designed in a CAD program and printed onto a high-resolution transparency or a conventional chrome mask. This transparency serves as a photomask in 1:1 contact photolithography, creating a master with a positive bas-relief of photoresist on a silicon wafer. The master, in turn, acts as a mold for PDMS. Liquid PDMS prepolymer is poured over the master, cured, and then peeled to form a PDMS replica. After plasma oxidation, the replica is sealed to a flat PDMS, glass, or silicon surface to

create the microfluidic channels. The entire process typically takes around 24 hours. Refer to Figure for a schematic representation of the involved procedures.

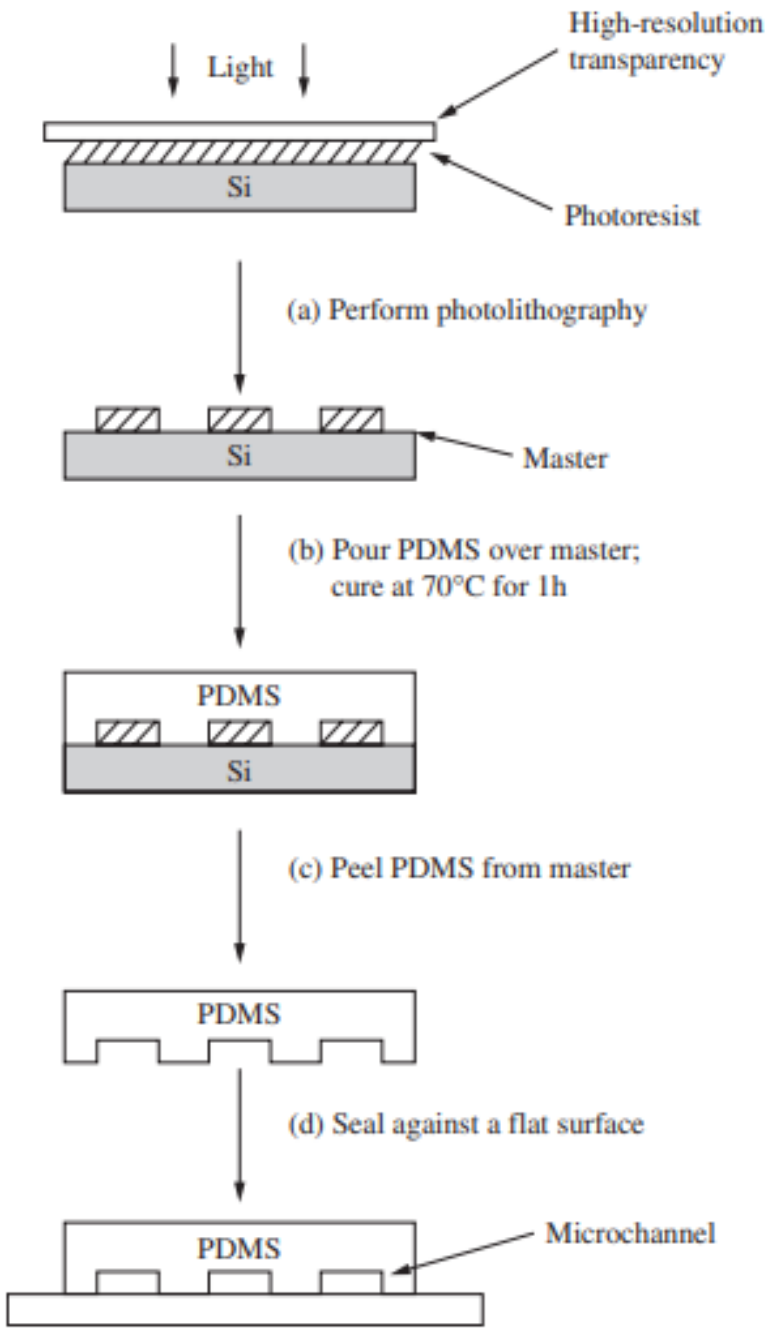

Figure 1:Scheme describing rapid prototyping of microfluidic systems

A system of channels is designed in a CAD program. A commercial printer uses the CAD fi le to produce a high-resolution transparency (~5000 dpi). (a) This transparency is used as a photomask in contact photolithography to produce a master. A master consists of a positive relief of photoresist on a silicon wafer and serves as a mold for PDMS. (b) Liquid PDMS prepolymer is poured over the master and cured for 1 h at 70°C. (c) The PDMS replica is peeled from the master. (d) The replica is sealed to a fl at surface to enclose the channels. The overall process takes ~24 h.

1.2 Problem Statement

This thesis reports the design and fabrication of microfluidics devices using soft lithography. How can different microfluidics devices should be designed based on required flow rate and diffusion time, making their CAD model in SolidWorks and then shaping that computer model into real one using the process of lithography.

CHAPTER 2 Literature Review

Microfluidic devices – miniaturized labs on a chip – manipulate fluids with exquisite precision, revolutionizing fields like medicine, drug discovery, and environmental monitoring. These tiny powerhouses offer efficiency, control, and portability, enabling faster analyses, lower costs, and even on-site environmental checks. Soft lithography, a key fabrication technique, makes them accessible and versatile.

While not directly related to microfluidic fabrication, the invention of the transistor in 1947 by **William Shockley, Walter Brattain, and John Bardeen** at Bell Labs laid the foundation for its future development. These three brilliant scientists created the first working transistor, a tiny semiconductor device that could amplify electrical signals. Their invention used germanium and point-contact technology to control current flow with unprecedented precision. The transistor opened the door to miniaturization in electronics, eventually enabling the development of smaller, more powerful chips crucial for microfluidic devices' control systems.

In 1955, **Jay Andrus'** patent for a photoengraving technique paved the way for precise microfluidic channel creation. Andrus, an engineer at Eastman Kodak, developed a method that used light-sensitive polymers to selectively remove material and leave behind intricate microchannels. The technique involved coating a substrate with a light-sensitive polymer, exposing it to light through a patterned mask, and then dissolving the unexposed areas, leaving behind the desired microchannel pattern. Photoengraving offered high resolution and accuracy for microfluidic channel fabrication, making it a cornerstone technique for years to come.

Richard Sweet's 1965 invention of a working inkjet printing device opened a new avenue for microfluidic fabrication. Richard Sweet, working at IBM, developed a system that ejected tiny droplets of ink onto a surface to create patterns. The device used piezoelectric crystals to generate precise pressure pulses, forcing ink droplets through microscopic nozzles onto the substrate. Inkjet printing offered the potential for rapid, low-cost microfluidic channel patterning. While initially used for other applications, its adaptability and scalability laid the groundwork for future microfluidic printing techniques.

The year 1979 witnessed two crucial breakthroughs that propelled microfluidics into the spotlight: first photolithography and second microscale gas chromatography system. While the photolithography technique itself existed earlier, 1979 marked its first commercial use in magnetic hard disk drive data storage devices. IBM led the charge in this application. Photolithography employs light-sensitive materials like photoresists to precisely pattern microchannels on substrates. Light exposure alters the resist's properties, allowing targeted etching or deposition, creating desired channel geometries. The commercial success of photolithography in data storage demonstrated its scalability, precision, and reliability for microfluidic fabrication. This paved the way for its widespread adoption in diverse fields like microchips, biosensors, and chemical microreactors.

Microscale Gas Chromatography System: Researchers at Stanford University, notably Terry S. Buchanan, pioneered this development. They miniaturized gas chromatography, a method for separating and analyzing gas mixtures, by fabricating microchannels on silicon wafers using photolithographic techniques. This miniaturization brought significant advantages in sample volume, analysis speed, and portability. The Stanford team's work showcased the potential of microfluidics for complex analytical applications. It highlighted the benefits of miniaturization for improved efficiency, reduced consumption, and faster analysis times, inspiring further advancements in microfluidic systems for chemical and

biological analysis. These two major developments in 1979 established microfluidics as a field with immense potential for scientific and technological advancements. By combining the precision of photolithography with the analytical power of miniaturized gas chromatography, they set the stage for the rapid growth and diversification of microfluidic applications in the decades to come.

In 1986, Charles Hull revolutionized the fabrication landscape with his invention of stereolithography (SLA). This additive manufacturing technique paved the way for intricate microfluidic channel creation in 3D. Charles Hull, an American engineer, envisioned a process for building objects directly from a digital model using light-activated polymerization. SLA employs a laser beam to scan and selectively solidify a vat of liquid resin, layer by layer, based on a 3D computer model. This allows for the creation of complex microfluidic channels with precise geometries and internal features. SLA offered several advantages for microfluidic fabrication like direct 3D fabrication, unmatched design freedom and material versatility. The impact of SLA on microfluidics was profound. It enabled researchers and engineers to design and create microfluidic devices with unprecedented complexity and functionality, opening doors for innovative applications in areas like microfluidics-based bioprinting, organ-on-a-chip models, and microfluidic heat exchangers.

While 1997 witnessed several pivotal advancements in microfluidics, the introduction of photoablation by **Stephen P. Roberts** and his team marked a significant milestone. This innovative technique harnessed the power of light to directly carve intricate microfluidic channels within polymer materials, offering a new avenue for microdevice fabrication. Stephen P. Roberts, along with colleagues at Sandia National Laboratories, recognized the potential of laser ablation for microfluidic channel creation. His team pioneered the technique and demonstrated its viability for various polymer materials. Photoablation utilizes focused laser beams to selectively remove material through various mechanisms. For polymers, the laser energy can ablate material through vaporization, decomposition, or photomechanical effects. By precisely controlling the laser parameters and scanning patterns, intricate microfluidic channel geometries can be sculpted directly within the polymer substrate.

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- Direct channel creation: Eliminated the need for intermediate masks or molds, simplifying the process and enabling rapid prototyping.
- High resolution and accuracy: Offered exceptional precision in channel dimensions and feature sizes, crucial for complex microfluidic circuits.
- Material versatility: Applicable to various polymers with diverse properties, catering to specific needs like biocompatibility or chemical resistance.

CHAPTER 3 Design Methodology

3.1 Problem Statement

Let's we have to mix benzene with water in a micromixer with two inlets at room temperature. The flow rates of both ethanol and water are 10 µL/min. Design the length of mixing channel if droplet size requirement is 100 µm.

Solution:-

According to Table 10.2 [6], the diffusion coefficient of ethanol in water at room temperature (25 °C) is $1.02 \times 10^{-5}\ cm^2/s$. Channel width will be equal to droplet size, 100 µm.

Solute	*D* ($\times 10^{-5} cm^2/s$)	*Solute*	*D* ($\times 10^{-5} cm^2/s$)
Air	2.00	Ammonia	1.64
Carbon dioxide	1.92	Benzene	1.02
Chlorine	1.25	Sulfuric acid	1.73
Ethane	1.20	Nitric acid	2.60
Ethylene	1.87	Acetylene	0.88
Hydrogen	4.50	Methanol	0.84
Methane	1.49	Ethanol	0.84
Nitrogen	1.88	Formic acid	1.50
Oxygen	2.10	Acetic acid	1.21
Propane	0.97	Propionic acid	1.06
Glycine	1.06	Benzoic acid	1.00
Valine	0.83	Acetone	1.16
Ovalbumin	0.078	Urease	0.035
Hemoglobin	0.069	Fibrinogen	0.020

Table 2: Table of Diffusion Coefficients in Water [8]

$$\mathrm{Fo} = \frac{\tau D}{w^2} \quad \Rightarrow \quad \tau = \mathrm{Fo}\frac{w^2}{D}$$

Assuming Fourier Number to be 0.5

Putting values in above equation:-

$$\tau = 0.5\ \frac{(100 \times 10^{-6})^2}{1.02 \times 10^{-5} \times 10^{-4}} = 4.90\ seconds$$

The average velocity of mixed liquid is:-

$$u = \frac{\dot{Q}_{water} + \dot{Q}_{benzene}}{A} = \frac{2 \times 10 \times 10^{-9}/60}{(100 \times 10^{-6})^2}$$

$$u = 33.33 \times 10^{-3}\ m/s$$

Thus the requires length of mixing channel is:-

$$L = u\tau = 33.33 \times 10 \times 4.90 = 163\ \mathrm{mm}$$

Total surface area required for the mixing channel is:-

$$A = 2wL = 2 \times 100 \times 163 \times 1000$$

$$A = 3.26 \times 10^7 \mu m^2$$

The dimension of the square area is:-

$$a = \sqrt{A} = \sqrt{3.26 \times 10^7} \approx 5709\ \mu m$$

Each turn consumes 2 × 100 µm for walls and 2 × 100 µm for channel width.

The total number of turns is:

$$N = \frac{5709}{4 \times 100} \approx 14$$

3.2 CAD MODEL

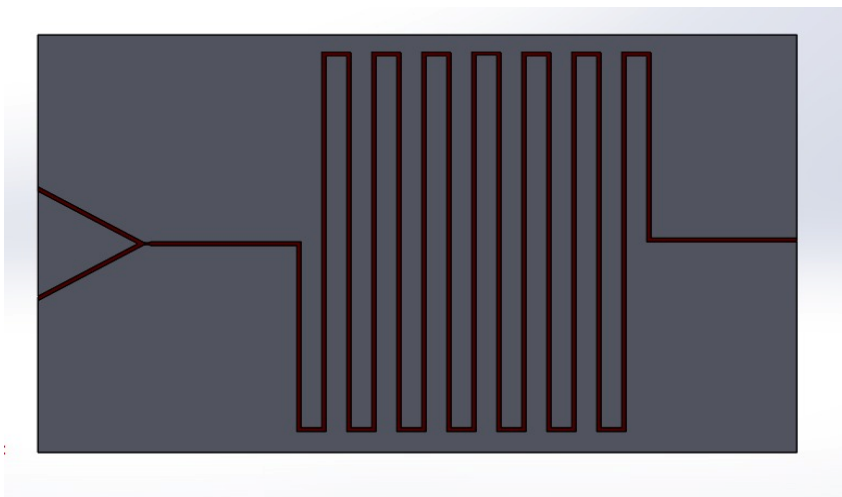

Figure 2: CAD Model

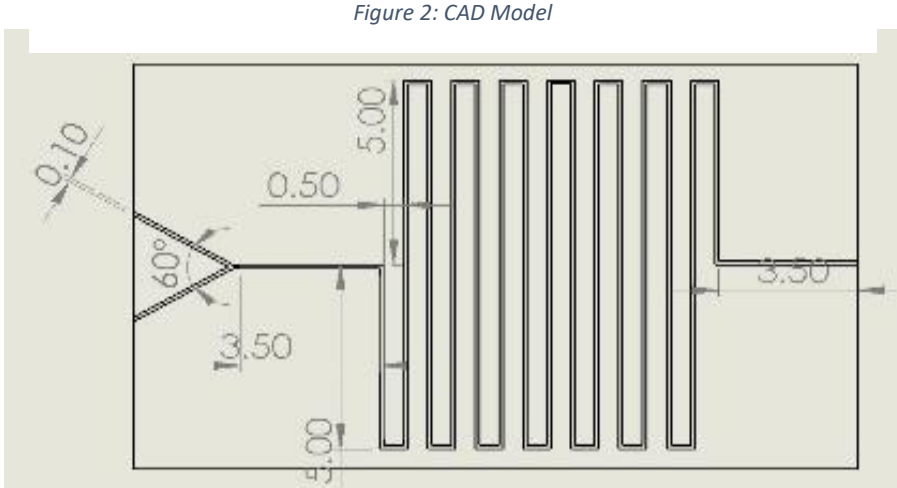

Figure 3: Engineering Drawing of Previous Model

3.3 Microfabrication Technique

Soft lithography is a versatile technique used in the fabrication of microfluidic devices. It involves the use of elastomeric materials, typically polydimethylsiloxane (PDMS), to replicate microscale features from a master mold onto the surface of the device. Here's an overview of how soft lithography is done:

Soft lithography is a versatile method widely utilized in the construction of microfluidic devices. It relies on elastomeric materials, such as polydimethylsiloxane (PDMS), to replicate microscale features onto the device surface. The initial phase involves creating a master mold with the desired microscale features, typically done through photolithography. This process includes coating a silicon wafer with a photosensitive material, exposing it to UV light through a photomask, and subsequently developing the exposed photoresist to unveil the microscale pattern on the silicon wafer.

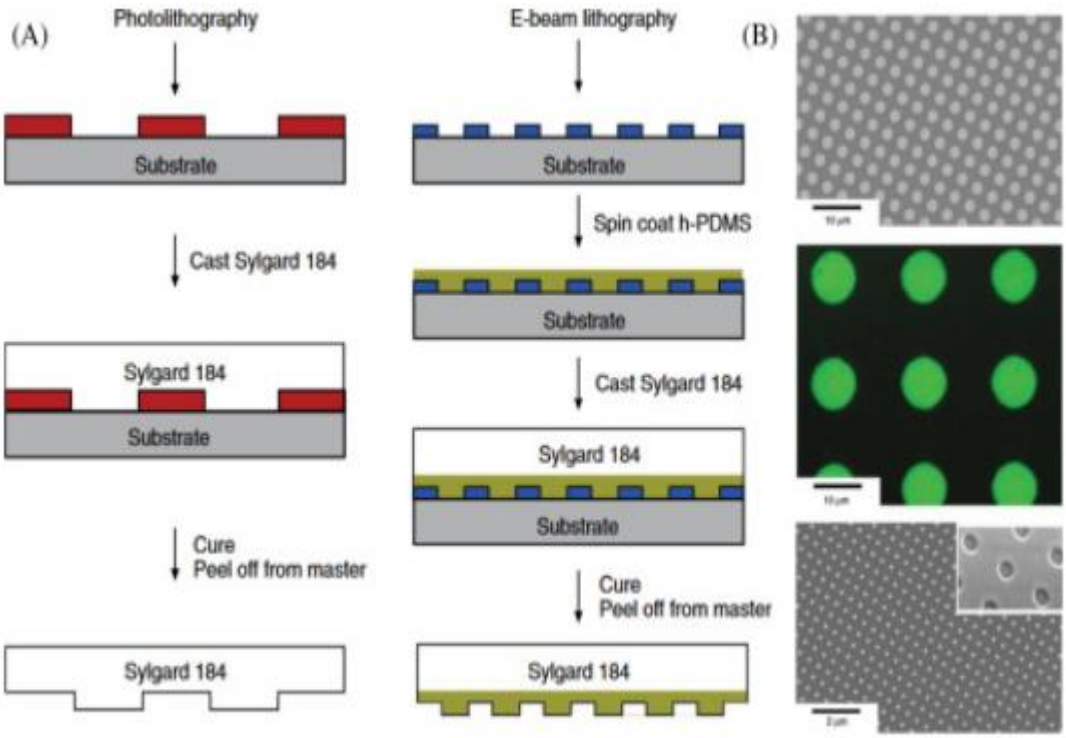

Figure 4: Overview of preparation of mold using Photolithography

After the fabrication of the master mold, the PDMS prepolymer is meticulously prepared by mixing the base polymer and curing agent in the specified ratio, often at a 10:1 ratio. Subsequently, the PDMS is cast onto the master mold to ensure uniform coverage. To eliminate air bubbles, the PDMS undergoes a degassing process within a vacuum chamber. The next step involves curing the PDMS through a heating process, promoting the crosslinking of polymer chains and solidification.

In the demolding process, the cured PDMS is delicately peeled from the master mold. This negative pattern of the master mold is now successfully replicated onto the PDMS surface. The subsequent steps, including punching holes for inlets and outlets, surface treatment with oxygen plasma, bonding with a substrate (often glass), and the testing and validation of the microfluidic device, ensure a functional and efficient final product. Soft lithography, with its adaptability and cost-effectiveness, has become a prominent method in the creation of intricate microfluidic devices for diverse applications.

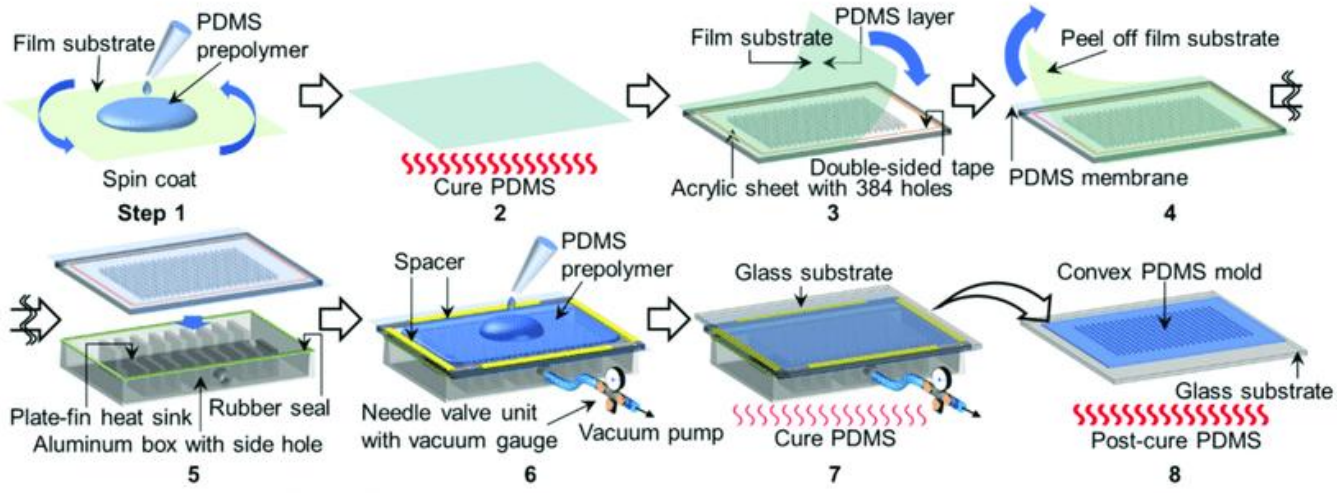

Figure 5: PDMS Casting Process

PDMS Polymer Mixing
Prepare the PDMS prepolymer by mixing the base polymer and curing agent in the appropriate ratio. Commonly, a 10:1 ratio is used.
PDMS Casting
Pour the mixed PDMS onto the master mold to cover the entire surface evenly. Remove air bubbles from the PDMS by placing it in a vacuum chamber. Cure the PDMS by heating it, causing the polymer chains to crosslink and solidify. This typically takes a few hours.
Demolding PDMS Replica:
Carefully peel the cured PDMS replica from the master mold. The negative pattern of the master mold is now replicated onto the surface of the PDMS.
Punching Inlets and Outlets:
Use a punch or a sharp tool to create holes at designated locations for inlets and outlets on the PDMS replica. These holes will later serve as entry and exit points for fluids.
Surface Treatment
Treat both the PDMS replica and another substrate (often glass) with oxygen plasma. This treatment enhances the bonding between the PDMS and the substrate.
Bonding PDMS to Substrate:
Bring the oxygen-plasma-treated surfaces of the PDMS replica and the substrate into contact, ensuring proper alignment. The interaction between the treated surfaces creates a strong and irreversible bond.
Curing and Sealing:

Heat the bonded PDMS and substrate ensemble to further enhance adhesion and to create a sealed microfluidic device.

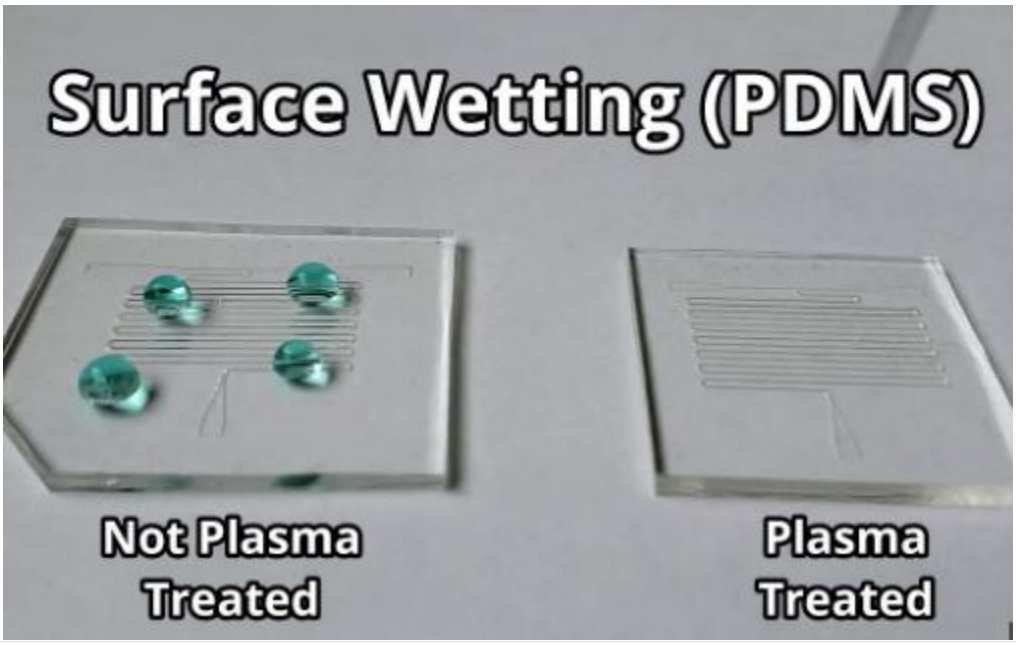

Figure 6: Surface Wetting (PDMS)

Connection Setup:

Integrate tubing into the punched inlet and outlet holes to allow for the introduction and removal of fluids.

Connect the microfluidic device to external components such as syringe pumps for fluid control.

Testing and Validation:

Perform tests to validate the functionality of the microfluidic device, ensuring that fluid flows through the channels as intended.

Check for leaks and verify the performance under various conditions.

Chapter 4: Fabrication Methodology

The fabrication of microfluidic devices involves a series of precise and controlled processes. This chapter details the methodologies we employ in the design and fabrication of microfluidic channels using soft lithography. The specific values provided here are those we use for fabricating a particular device; users should adjust these values based on their own requirements. This process is divided into two main parts:

1. Preparation of the mold via soft photolithography
2. Casting of PDMS channels using the prepared mold.

4.1 Preparation of Mold

To begin with, it is essential to ensure that the substrate surface is hydrophilic. While silicon wafers are preferred due to their superior properties, their high cost often necessitates the use of glass slides as a substrate. In our initial experiments, glass slides were used to become familiar with the lithography process before transitioning to silicon wafers.

4.1.1 Argon Plasma Treatment

To render the glass slides hydrophilic, argon plasma treatment was employed. This treatment increases the surface energy of the substrate, thereby enhancing the adhesion between the glass slide and the SU-8 photoresist. The process involves:

1. **Cleaning the Substrate**: Initially, the glass slides are thoroughly cleaned using isopropyl alcohol (IPA) and deionized (DI) water to remove any organic contaminants.

2. **Plasma Treatment**: The cleaned glass slides are placed in a plasma cleaner. Argon gas is introduced, and the slides are exposed to the plasma for 10 minutes. The plasma treatment increases surface energy by adding functional groups, such as hydroxyl groups, making the surface hydrophilic.

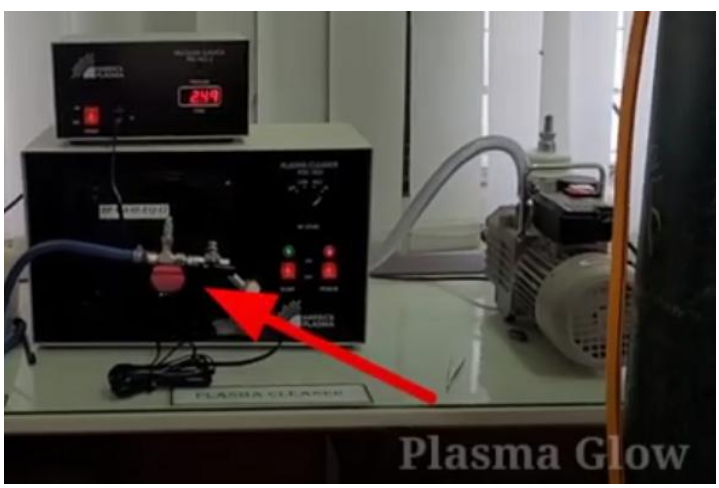

Figure 7: Plasma Treatment

The hydrophilicity of the treated substrate can be assessed using the contact angle of a water droplet. A flat droplet indicates a hydrophilic surface, while a dome-shaped droplet indicates hydrophobicity. The contact angle was measured by placing a water droplet on both treated and untreated glass slides and observing the shape and spread of the droplet. A contact angle goniometer was used for precise measurement. The treated slides showed a significant decrease in contact angle compared to the untreated slides, confirming the effectiveness of the plasma treatment.

Figure 8: Water droplet on a plasma treated surface

Figure 9: Water droplet on an untreated surface

4.1.2 Spin Coating with Photoresist

The next step involves spin coating the substrate with SU-8 2075 photoresist to achieve a desired thickness of 200 microns. The spin coating process ensures a uniform layer of photoresist on the substrate. This technique is crucial for achieving the precise thickness and uniformity required for subsequent lithographic processes. Proper control of the spin speed and duration is essential to reach the target thickness, followed by a soft bake to solidify the photoresist layer before further processing.

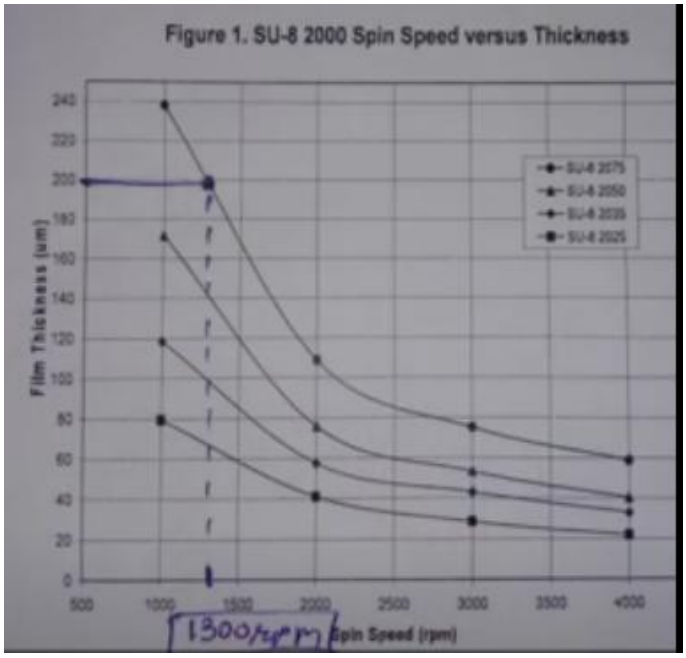

Figure 10: Extracting value of speed against desired thickness

The spin coater was set to 1300 RPM based on data provided in the photoresist manufacturer's datasheet. The procedure involves:

1. **Preparation**: The hydrophilic glass slide is placed on the spin coater chuck and secured with a vacuum.
2. **Photoresist Application**: Approximately 5-10 ml of SU-8 2075 photoresist is deposited onto the center of the glass slide.
3. **Spin Coating**: The spin coater is programmed to spin the substrate at 1300 RPM for 30 seconds. This speed and duration are critical to achieving a uniform 200-micron thick layer. The centrifugal force spreads the photoresist evenly across the substrate.
4. **Edge Bead Removal**: Excess photoresist at the edges can be removed using a cleanroom wipe moistened with acetone or IPA.

Figure 11 : Coating substrate with SU-8 photoresist

4.1.3 Soft Baking

Soft baking is crucial for evaporating the solvent from the photoresist, thus improving adhesion between the photoresist and the substrate and allowing the UV mask to be applied without sticking. This step typically involves heating the coated substrate on a hotplate or in an oven at a specific temperature and duration, depending on the photoresist formulation. Proper soft baking ensures the photoresist achieves the necessary hardness and reduces the risk of film cracking or delamination during subsequent processing steps. It also helps in achieving a more defined and accurate pattern during the UV exposure step.

Figure 12 : Soft Baking

Based on the SU-8 datasheet, the substrate was soft-baked in a convection oven with the following parameters:

THICKNESS	SOFT BAKE TIMES	
	(65°C)	(95°C)
microns	minutes	minutes
25 -40	0 - 3	5 - 6
45 - 80	0 - 3	6 - 9
85 - 110	5	10 - 20
115 - 150	5	20 - 30
160 - 225	7	30 - 45

Table 3 : Soft Baking Time Duration for Required Thickness

1. **Initial Baking**: The coated substrate is placed in an oven at 65°C for 7 minutes. This step initiates solvent evaporation and partially hardens the photoresist.
2. **Secondary Baking**: The temperature is then increased to 95°C and baked for an additional 35 minutes. This step ensures complete solvent evaporation, which is critical for the subsequent UV exposure.

3. **Cooling**: The substrate is allowed to cool gradually to room temperature to avoid thermal shock and potential warping.

4.1.4 UV Exposure

For UV exposure, a photomask with the desired design was placed onto the photoresist-coated substrate. The UV exposure process transfers the design from the photomask to the photoresist.

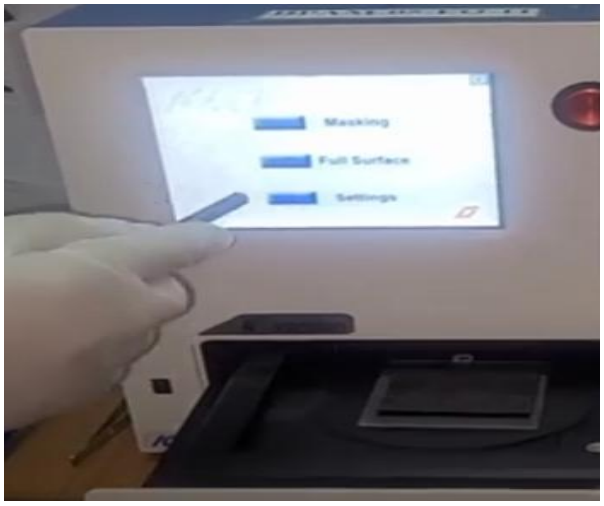

Figure 13 : UV Exposure

The photomask design was created using CAD software such as SolidWorks. The design was printed onto a transparent film suitable for UV exposure. The resolution of the design and the quality of the photomask are critical for the precision of the microfluidic channels.

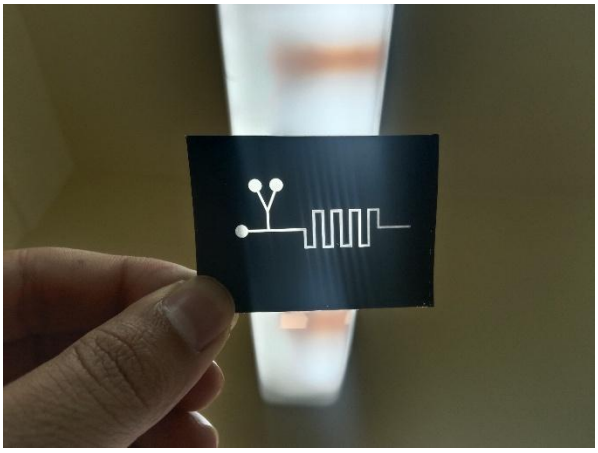

Figure 14 : Photomask

The appropriate UV exposure dosage, as recommended in the datasheet, was 45 seconds for a 200-micron thickness. The procedure involves:

1. **Alignment**: The photomask is carefully aligned with the photoresist-coated substrate using an aligner tool to ensure precise pattern transfer.

2. **UV Exposure**: The aligned photomask and substrate are placed in a UV exposure system. UV light is shone through the photomask for 45 seconds, initiating a photochemical reaction in the exposed areas of the photoresist.

3. **Post-Exposure Alignment Check**: After exposure, the photomask is carefully removed, and the substrate is inspected under a microscope to ensure the design has been accurately transferred.

THICKNESS microns	EXPOSURE ENERGY mJ/cm^2
25 - 40	150 - 160
45 - 80	150 - 215
85 - 110	215 - 240
115 - 150	240 - 260
160 - 225	260 - 350

Table 4 : UV Exposure Energy against required thickness

4.1.5 Post-Exposure Baking

Post-exposure baking further evaporates solvent, enhances adhesion, and reduces stress by promoting mechanical relaxation. This step is crucial for improving the structural integrity and dimensional stability of the photoresist patterns, ensuring they remain intact during subsequent processing stages.

According to the SU-8 datasheet, the substrate was baked with the following parameters:

1. **Initial Bake**: The substrate is baked at 65°C for 5 minutes to further solidify the exposed photoresist.

THICKNESS microns	PEB TIME (65^oC)* minutes	PEB TIME (95^oC) minutes
25 -40	1	5 - 6
45 - 80	1 - 2	6 - 7
85 - 110	2 - 5	8 - 10
115 - 150	5	10 - 12
160 - 225	5	12 - 15

Table 5: Post Exposure Baking Time

2. **Secondary Bake**: The temperature is then increased to 95°C and baked for an additional 15 minutes. This step strengthens the cross-linked network formed during UV exposure.

3. **Cooling**: The substrate is allowed to cool to room temperature.

Figure 15 : Placing substrate in oven for PEB

4.1.6 Pattern Visibility Check

A visible pattern after post-exposure baking indicates sufficient UV exposure. This can be checked by inspecting the substrate under a microscope to ensure the desired design is visible and well-defined. Any discrepancies or defects at this stage need to be addressed before proceeding to ensure the quality and accuracy of the final microfluidic channels

Figure 16: Pattern Visibility Check

4.1.7 Developing the Channel

The developed channel is created using SU-8 developer. The developer selectively removes the unexposed photoresist, leaving behind the patterned microchannels. This process involves immersing the substrate in the developer solution and gently agitating it to ensure complete removal of the unwanted photoresist. The development time and conditions must be carefully controlled to achieve precise and clean channel patterns. After development, the substrate is typically rinsed with isopropyl alcohol (IPA) to stop the development process and remove any residual developer, followed by a final hard bake to further solidify and stabilize the remaining photoresist structures.

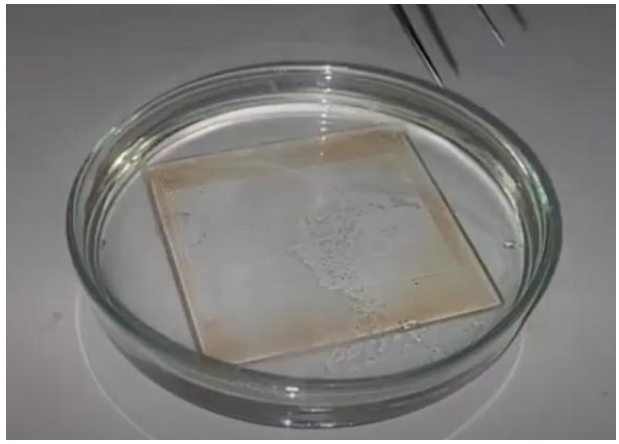

Figure 17 : Developing the Channel

The development process involves:

1. **Immersion**: The baked substrate is submerged into a beaker containing SU-8 developer.
2. **Agitation**: The beaker is gently agitated or placed on a magnetic stirrer for 15 minutes to ensure uniform development.
3. **Rinsing**: After development, the substrate is rinsed with isopropyl alcohol (IPA) to remove any residual developer. The substrate is then briefly rinsed with DI water.
4. **Drying**: The substrate is dried using an air compressor to achieve a clean and well-defined microchannel pattern. It is critical to avoid drying marks, so a filtered air source is recommended.

THICKNESS microns	DEVELOPMENT TIME minutes
25 - 40	4 - 5
45 - 75	5 - 7
80 - 110	7 - 10
115 - 150	10 - 15
160 -225	15 - 17

Table 6 : Development time against required thickness

4.2 PDMS Casting

The second phase involves PDMS casting to create flexible microfluidic channels.

4.2.1 Mixing PDMS Elastomer and Curing Agent

The PDMS elastomer and curing agent were mixed in a 10:1 ratio. The procedure involves:

1. **Weighing**: A beaker is placed on the weight balance and set to zero. 52 grams of PDMS elastomer are added to the beaker.

Figure 18 : Weighing PDMS Elastomer

2. **Adding Curing Agent**: 5 grams of curing agent are then added to the beaker.

Figure 19 : PDMS+Curing Agent weight

3. **Mixing**: The elastomer and curing agent are rigorously mixed using a stirring rod or mechanical mixer to form a homogeneous solution. Proper mixing is essential to ensure uniform curing.

4.2.2 Degassing the Mixture

The mixture contains bubbles that need to be removed to ensure a clear PDMS cast.

Figure 20: Mixture Containing Bubbles

The degassing process involves:

1. **Vacuum Desiccator**: The mixed PDMS solution is placed in a vacuum desiccator to remove air bubbles. This process can take up to 30 minutes, depending on the volume of the mixture and the efficiency of the vacuum pump.

Figure 21 : Degassing through Vacuum Desiccator

2. **Vacuum Chamber**: To expedite degassing, a chamber with a vacuum pump was used. The low pressure in the chamber allows bubbles to rise and burst more quickly. The degassed PDMS solution becomes clear and bubble-free, ready for casting.

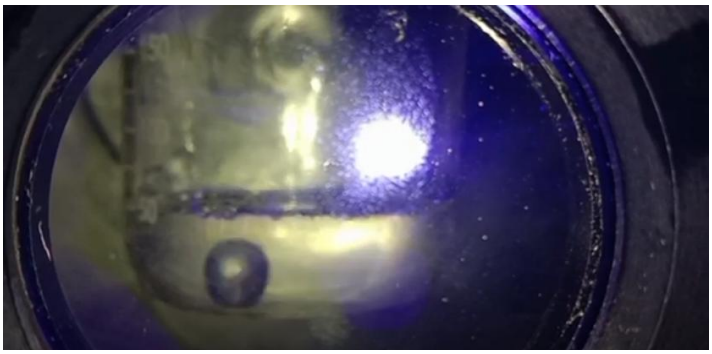

Figure 22 : Vacuum Chamber

4.2.3 PDMS Casting and Curing

The PDMS casting process involves:

1. **Pouring**: The degassed PDMS solution is carefully poured over the prepared mold to cover the entire surface evenly.

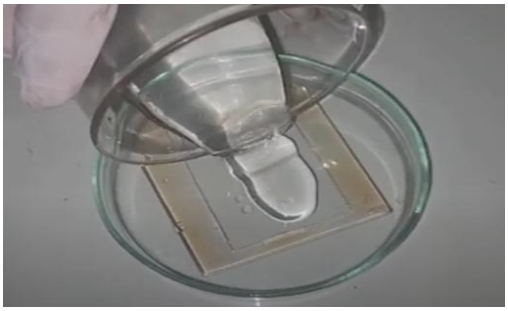

Figure 23 : Pouring PDMS

2. **Curing**: The mold is placed in an oven at 65°C for 3 to 4 hours. This step ensures that the PDMS cures and solidifies, forming the microfluidic channels.

Figure 24 : Curing PDMS

3. **Cooling and Demolding**: After curing, the mold is allowed to cool to room temperature. The cured PDMS is then carefully peeled off from the mold.

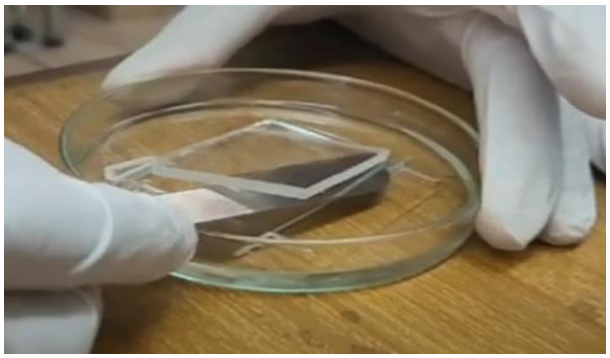

Figure 25: Peeling of f the PDMS

4.2.4 Punching Holes

Inlet and outlet holes are punched into the PDMS layer using a biopsy punch or similar tool. The placement and size of these holes are critical for fluid flow control in the microfluidic device. Precise alignment ensures efficient fluid entry and exit, minimizing the risk of leaks and ensuring the device functions as intended.

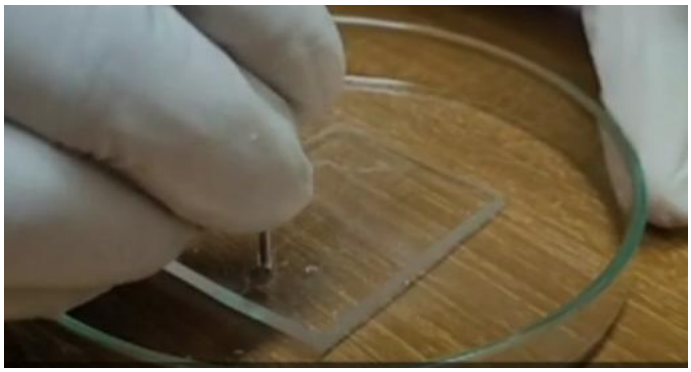

Figure 26: Punching Holes

4.2.5 Plasma Bonding

The PDMS layer and another substrate (either another PDMS layer or a glass slide) are plasma treated to enhance bonding. The process involves:

- **Plasma Treatment**: Both surfaces are treated in a plasma cleaner for 1 to 1.5 minutes, depending on the plasma gas used. This treatment activates the surfaces by generating reactive oxygen species, which increases their surface energy.

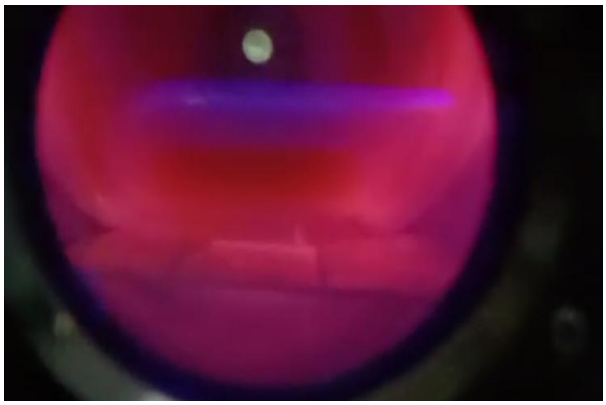

Figure 27 : Plasma Bonding

- **Bonding**: Immediately after plasma treatment, the treated surfaces are brought into contact and pressed together firmly. Proper alignment is crucial to ensure the channels are sealed correctly.

- **Bond Strengthening**: Although the initial bond is weak, leaving the bonded device at 60°C for 3 days significantly strengthens the bond. This step ensures that the bond between the PDMS layers or between PDMS and the glass slide is robust and durable.

4.3 Validation and Testing

The final fabricated microfluidic device must undergo validation and testing to ensure functionality and reliability.

1. **Visual Inspection**: The bonded device is inspected under a microscope to check for any defects, misalignments, or incomplete bonding.
2. **Leak Test**: The device is subjected to a leak test by introducing a fluid (usually colored dye) through the inlet and monitoring for any leaks or blockages in the channels.
3. **Functional Testing**: The device is tested under operational conditions to verify that it meets the required specifications for fluid flow, mixing, or other intended functionalities.

4.4 Conclusion

The fabrication of microfluidic devices using soft lithography involves multiple precise and controlled steps. Each step, from substrate preparation to PDMS casting and bonding, requires careful attention to detail to ensure the successful creation of functional microfluidic channels. This chapter has detailed the methodologies and procedures followed in the fabrication process, providing a comprehensive guide for future fabrication efforts.

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